System Design

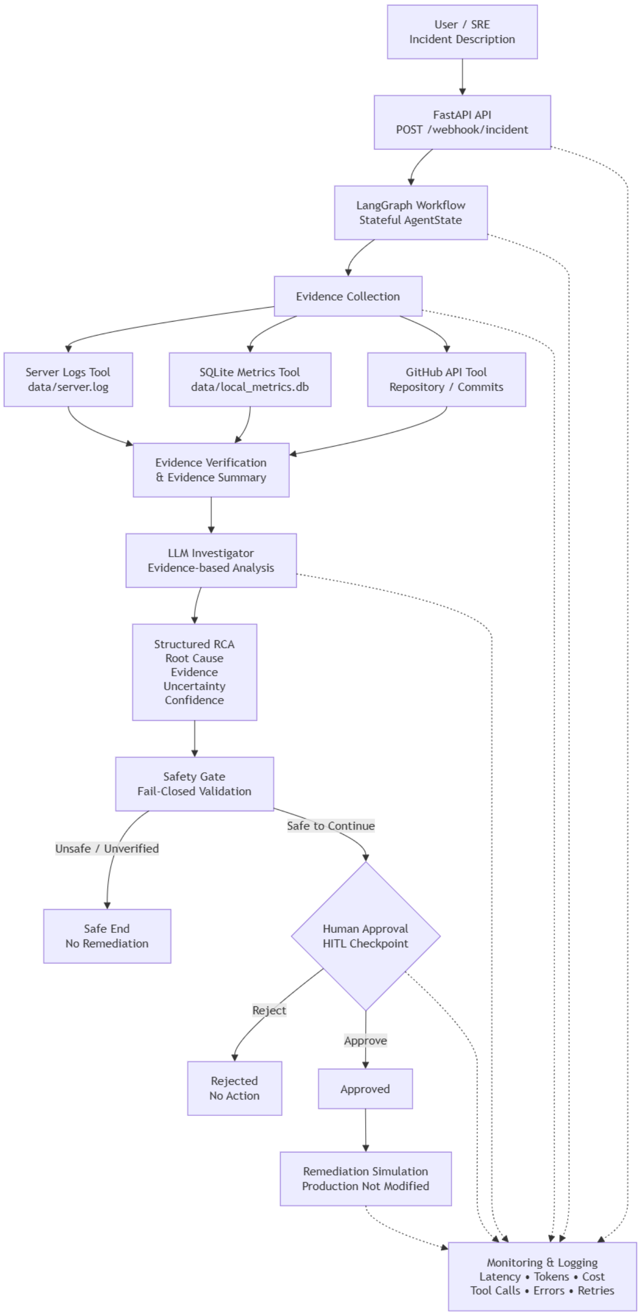
SRE Incident Investigation & Safe Remediation Agent

Use Case

The proposed system is an AI-powered SRE incident investigation agent. It receives an incident through a FastAPI endpoint, gathers evidence from server logs, local infrastructure metrics, and GitHub repository activity, and uses an LLM to produce an evidence-based root-cause analysis (RCA). Before any consequential remediation action, the workflow pauses for explicit human approval. The current remediation path is simulation-only and does not modify production infrastructure.

Business Goal

The goal is to reduce the time and inconsistency involved in initial incident investigation while preserving operational safety. The agent automates evidence collection and RCA generation, but deterministic application logic controls validation, safety checks, human approval, and remediation state.



System Components

Component	Responsibility
FastAPI	Receives incident requests and human approval decisions.
LangGraph	Orchestrates the stateful investigation and approval workflow.
AgentState	Maintains incident, evidence, RCA, approval and remediation state.
Server Log Tool	Retrieves service-specific application/server logs.
SQLite Metrics Tool	Retrieves CPU, memory and database connection metrics.
GitHub Tool	Retrieves the latest repository commit and changed files.
LLM Investigator	Analyzes collected evidence and produces an evidence-based investigation.
RCA Generator	Structures root cause, evidence, uncertainty, conflicts and confidence.
Safety Gate	Prevents unsafe or unverified remediation paths.
HITL Checkpoint	Requires explicit human approval before consequential remediation.
Monitoring	Records latency, token usage, cost, errors, retries and tool calls.

Data Sources and Tools

The agent uses three evidence sources. The server-log tool reads the local application log file and filters entries by service. The SQLite tool queries service-specific CPU, memory and database-connection metrics using a parameterized SQL query. The GitHub tool calls the GitHub REST API to retrieve the latest commit and changed files for the configured repository. Tool failures are handled through timeouts, retries and fallback responses.

Agent State

Incident state: incident_id, service_name, repository, description.
Evidence state: logs, metrics, commits, service verification and evidence summary.
RCA state: root cause, evidence summary, conflicting evidence, uncertainty and confidence.
Safety/HITL state: approval_required, approval_status, human_approved.
Remediation state: remediation_action, remediation_status, remediation_executed and production_modified.

Human-in-the-Loop Checkpoint

The workflow uses a LangGraph interrupt to pause execution before remediation. The human receives an explicit approve/reject decision. Rejection terminates the remediation path, while approval allows the controlled remediation simulation. The system tracks approval and remediation state explicitly and does not claim that production was modified unless the state says so.

Framework Choice and Rationale

We selected LangGraph because this SRE incident investigation problem is a stateful, control-heavy workflow requiring deterministic transitions, evidence collection, safety gates, and human approval. LangGraph allows the system to maintain structured incident state across investigation stages and pause execution at a human-in-the-loop checkpoint before consequential actions. CrewAI was not selected because the primary requirement is workflow control and safe state transitions rather than collaboration between multiple autonomous roles. The LLM is therefore used for evidence-based investigation and RCA generation, while deterministic application logic controls validation, safety, approval, and remediation.

Safety Design

The design follows a fail-closed approach. The LLM does not directly execute production operations. Deterministic safety logic evaluates verification and approval state, and the human checkpoint is placed before the remediation stage. The current implementation keeps remediation simulation-only, with production modification explicitly represented as false.